NURTURING INNOVATIONS

TIFAC has not restricted itself in predicting future only, it has also played an important role in evolution of the innovation ecosystem (ideas to commercialisation) in the country by acting as a test-bed. Some notable examples include Home Grown Technology (HGT) programme which introduced the concept of supporting the scaling up efforts of indigenous technology through academia-industry collaborations.; Technopreneur Promotion Programme for the first time in the country supported individual innovators to develop prototypes, and SRIJAN programme has scaled up technology innovation towards commercialisation. Such innovation programmes have contributed to Indian economy through revenue generation.

EMPOWERING WOMEN

TIFAC has played a pioneering role in creating an enabling environment in the domain of Intellectual Property Rights (IPR) i.e. IP filing and management and capacity building. Similarly, Women Scientist Scheme-C, WOS-C or WISE-KIRAN IPR programme empowers women, who had to take a break mid-career due to any reason, to return to the mainstream science. It is not only empowering the women but also creating human resources for strengthening the S&T ecosystem of the country."By this scheme women are being encouraged to pursue research in frontier areas of science and engineering, on problems of societal relevance and to take up S&T-based internship followed by self-employment", says Reena MP Singh, 7th batch trainee, now a Patent Associate with Kan and Krishme, New Delhi. This scheme bagged Nari Shakti Award from Hon'ble President of India in 2015.

HANDHOLDING OF MSMEs

TIFAC has also been providing R&D and technical support to technologically homogenous MSME clusters to nurture technology infusion and self-reliance. So far more than 45 MSME clusters have been taken up under the programme wherein Technology Gap Analysis





"TIFAC, serving as Technology think tank of India, identifies, evaluates, accesses and demonstrates technology innovation as a tool for societal and economic development of our nation towards aatmanirbharta and bringing transformative changes."

— Dr Pradeep Srivastava, Executive Director, TIFAC

Studies have been carried out and various soft and hard interventions have been made with the help of proximate academia/R&D and technical institutions.

QUICK RESPONSE TO COVID-19 PANDEMIC

During the Covid-19 pandemic period, TIFAC expanded its foresight capability by way of various initiatives such as white paper, policy paper, action plan documents, demonstration of select technologies for the benefit of the society towards bringing back Indian economy on track and achieving Aatmanirbharta. These reports are well accepted in Government system.

JOB CREATION FOR SHRAMIKS

TIFAC set up Shramik Shakti Manch (SAKSHAM), dynamic app / portal for mapping skills of migrant labourers vis-a-vis industry requirements. This timely and unique initiative has been a huge success towards providing gainful employment and at the same time creating a database of labourers with their skill sets which is helping MSMEs and other industries fulfilling their requirements. Prateek Yashwant Katker, a resident of Mumbai could bag job as computer operator in Priyanka Marbles through this platform. Another unemployed youth, Vijay Sravan of Mumbai is now employed as office boy at a stone marble company in Maharashtra. So far thousands of MSME industries and lakhs of shramiks have already registered on SAKSHAM.

EXTENDING HEALTHCARE TO VILLAGES

Inspired by Prime Minister Narendra Modi's vision of "SabkaSwasthya, SabkaVikas", TIFAC has conceptualised a scalable pilot demonstration telemedicine project (Plug and Play model) to provide quality medical care to underprivileged women and children in remote areas of Varanasi, Gorakhpur and Kamjong (Manipur) at an affordable cost where very skewed doctor-patient ratio could not deliver healthcare facilities. Just with a single blood drop, several tests like blood glucose level, lipid profile and haemoglobin and various other basic health parameters electrocardiographic examination (heart check-up), measurement of heart rate, BP, body temperature, blood oxygen content, and mother & child care (foetal doppler) are being measured and data can be transferred to any doctor online anywhere in the world. In this process, e-health records of citizens are also being generated.

WORKING FOR A GREENER FUTURE

TIFAC has been contributing immensely for assessment of technology needs towards attaining India's Nationally Determined Contributions (NDC) targets. In this direction, TIFAC is preparing decarbonising roadmap for industries in sync with emissions reduction targets set forth by Hon'ble PM of India at the UNFCCC (COP26) conference held in Glasgow in 2021.

Similarly, in association with the International Transport Forum at OECD (Organization for Economic Co-operation and Development), TIFAC has taken up a project on the assessment of real-life fuel economy, mileage and load/ passenger capacity for various modes and segments of transport under Indian conditions. To address the need of a bottom-up estimation of future penetration of electric two-wheelers on yearly-basis, TIFAC and NITI Aayog have jointly developed an agent-based tool.

ASSESSMENT OF TECHNOLOGY MATURITY FOR AATMANIRBHARTA (ATMA)

TIFAC has launched the ATMA programme which has extended handholding support to innovators for making business plans along with market potential mapping, scouting, obtaining regulatory approvals, and conducting pilot/field testing, etc. towards making such technologies for industry support. The technology certification proposed to be provided by TIFAC would be the first of its kind in Indian S&T and innovation system.

Thus, TIFAC has been playing a pivotal rule in foreseeing technology trends, supporting innovation, assessing their relevance for India in different sectors and demonstrating them through actions on ground. The organisation indeed has been a pioneer in mapping future technology trajectories for the nation. All such efforts are targeted to make India Aatmanirbhar in science and technology.